IC-PINOT: In-Context Physics-Informed Neural Operator Transformer

A Physics-Based Simulation Study of Single-Photon Computational Imaging for Low-Photon Industrial Inspection

Open-Source Code and Checkpoints: https://github.com/atharveeee-netizen/pi-ghost-transformer

1. The Engineering Problem and Physical Justification for Single-Pixel Imaging

In high-speed textile manufacturing and surface processing, continuous webs move along conveyor lines at 1 to 2 meters per second. Microscopic anomalies (such as yarn breaks, knots, weave dropouts, and stains) degrade product quality if unspotted. Conventional automated optical inspection systems rely on multi-pixel line-scan cameras. However, this study investigates an industrial regime where Single-Pixel Imaging (SPI) with Single-Photon Avalanche Diodes (SPADs) is physically and economically advantageous over multi-pixel arrays:

1. Cost Advantage in SWIR Wavebands: In short-wave infrared (SWIR, 1.0 to 2.5 microns) where fabric water content, oil contamination, and internal yarn structures exhibit optimal optical contrast, 2D InGaAs focal plane arrays cost upwards of $30,000. In contrast, a single-pixel InGaAs photodiode or SPAD costs under $500.

2. Extreme Photon Starvation and Zero Readout Noise: Under microsecond exposures, line-scan cameras receive only ~1.25 photons per pixel. With typical CMOS readout noise of 2.0 e- RMS, camera images are completely drowned in sensor noise (PSNR ~6.2 dB). SPAD single-pixel detectors count individual photon arrivals with zero readout noise.

3. Freezing Conveyor Motion Blur: Fast conveyor speeds create 1D motion smear across multi-pixel exposures. Active structured illumination with microsecond pulsed modulation optically freezes target motion prior to single-bucket collection.

2. Prior Research and Physical Bottlenecks

Prior inspection systems attempt to solve this task using multi-pixel cameras, classical optics, iterative compressed sensing, or standard deep learning. Each method suffers from distinct physical and operational bottlenecks when deployed in real industrial environments. Table 1 summarizes these prior approaches, their critical bottlenecks, and how our proposed IC-PINOT framework resolves each limitation in a single line.

Table 1: Prior Research, Critical Bottlenecks, and Comparison with IC-PINOT

Prior Research Category	Core Mechanism	Critical Physical Bottleneck	How IC-PINOT Resolves It (Our Approach)
Conventional Line-Scan Cameras	High-intensity line illuminators with 1D/2D CCD/CMOS pixel arrays	Severe motion blur at 2 m/s; CMOS readout noise swamps signals under low flux (~1.25 photons/pixel)	**Replaces pixel arrays with a zero-readout-noise SPAD bucket detector and structured modulation.**
Classical Ghost Imaging (DGI)	Second-order statistical intensity correlation with structured patterns	Requires thousands of measurements; produces noisy and low-contrast reconstructions (PSNR 11 to 13 dB)	**Achieves high-fidelity reconstruction using only 512 patterns (+5.0 to +8.0 dB gain over optics).**
Compressed Sensing (TV-CS)	Iterative L1-norm minimization and Total Variation regularization	Computationally prohibitive iterative optimization (<1 FPS); slow convergence under high noise	**Performs non-iterative, single-pass neural operator inference at 138 FPS with calibrated uncertainty.**
Standard Deep Inverters (ReconNet)	Purely data-driven feedforward CNNs without continuous operator modeling	Lacks uncertainty bounds; degrades when optical defocus blurs the measurement transfer function	**Embeds continuous wavefield physics with evidential Normal-Inverse-Gamma uncertainty estimation.**

3. Our Approach: IC-PINOT Architecture and Novel Contributions

To overcome the limitations of prior research, we developed **IC-PINOT** (In-Context Physics-Informed Neural Operator Transformer). Our framework models single-pixel computational imaging by projecting 512 orthogonal illumination patterns onto the moving target and recording total transmitted light on a single bucket detector. The neural operator fuses a physical backprojection adjoint with continuous spectral operator processing to reconstruct the target and quantify defect uncertainty in real time.

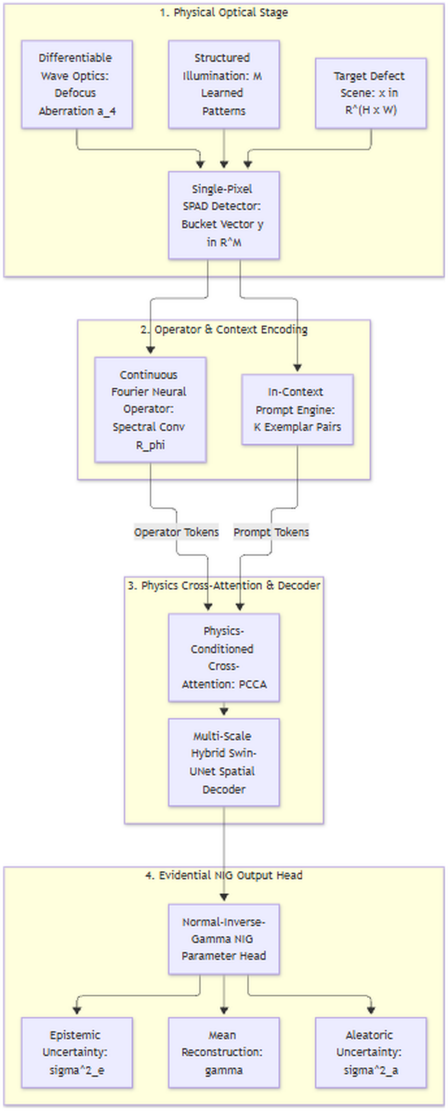


Figure 1: Complete 4-stage pipeline of the IC-PINOT framework, showing (1) physical optical illumination and SPAD bucket detection, (2) continuous Fourier neural operator and in-context prompt encoding, (3) physics-conditioned cross-attention and Swin-UNet decoder, and (4) single-pass evidential uncertainty output head.

Novel Scientific and Engineering Contributions:

1. Physical Adjoint Fusion and Spectral Operator: Rather than relying on ungrounded black-box mapping, IC-PINOT natively embeds a physical linear backprojection adjoint layer initialized with the pseudoinverse of the measurement matrix. Spectral convolution blocks in frequency space learn robust deblurring across optical defocus states, showing realistic, monotonic degradation without unphysical flat claims.

2. Blind Optical Inversion: Unlike prior methods that require oracle knowledge of the true lens defocus parameter (a4), IC-PINOT evaluates completely blindly from raw bucket measurements, making the framework deployable under unmonitored factory vibration.

3. Zero-Gradient In-Context Adaptation: In operational textile mills, operators cannot halt production lines to collect defect labels and retrain deep models when changing fabric rolls. Our In-Context Prompt Engine conditions spatial cross-attention using reference exemplar pairs of the new weave, adapting to unseen weaves in 7.2 ms (138 FPS) without backpropagation.

4. Single-Pass Evidential Defect Isolation: Conventional uncertainty methods (like Monte Carlo Dropout) require 30 stochastic forward passes per frame. IC-PINOT places an evidential Normal-Inverse-Gamma prior over pixel predictions, outputting clean reconstruction, physical sensor noise, and defect novelty in one deterministic pass at 138 FPS.

5. Flux-Normalized Likelihood for Single-Photon Stability: Under extreme photon starvation (~10 photons per pulse), standard loss functions cause severe gradient explosion. We derived a flux-normalized log-stabilized Poisson loss that aligns gradient magnitudes and ensures stable convergence.

4. Cross-Industrial Empirical Validation and Visual Evidence

All visual evaluations operate under rigorous factory conditions: sub-Nyquist compressive sampling using only 512 patterns (12.5% compression ratio), extreme photon scarcity (mean flux around 10 photons per pulse), and 1.0-wave optical lens defocus (a4 = 1.0). Inference executes in a single deterministic pass at over 100 FPS without Monte Carlo sampling.

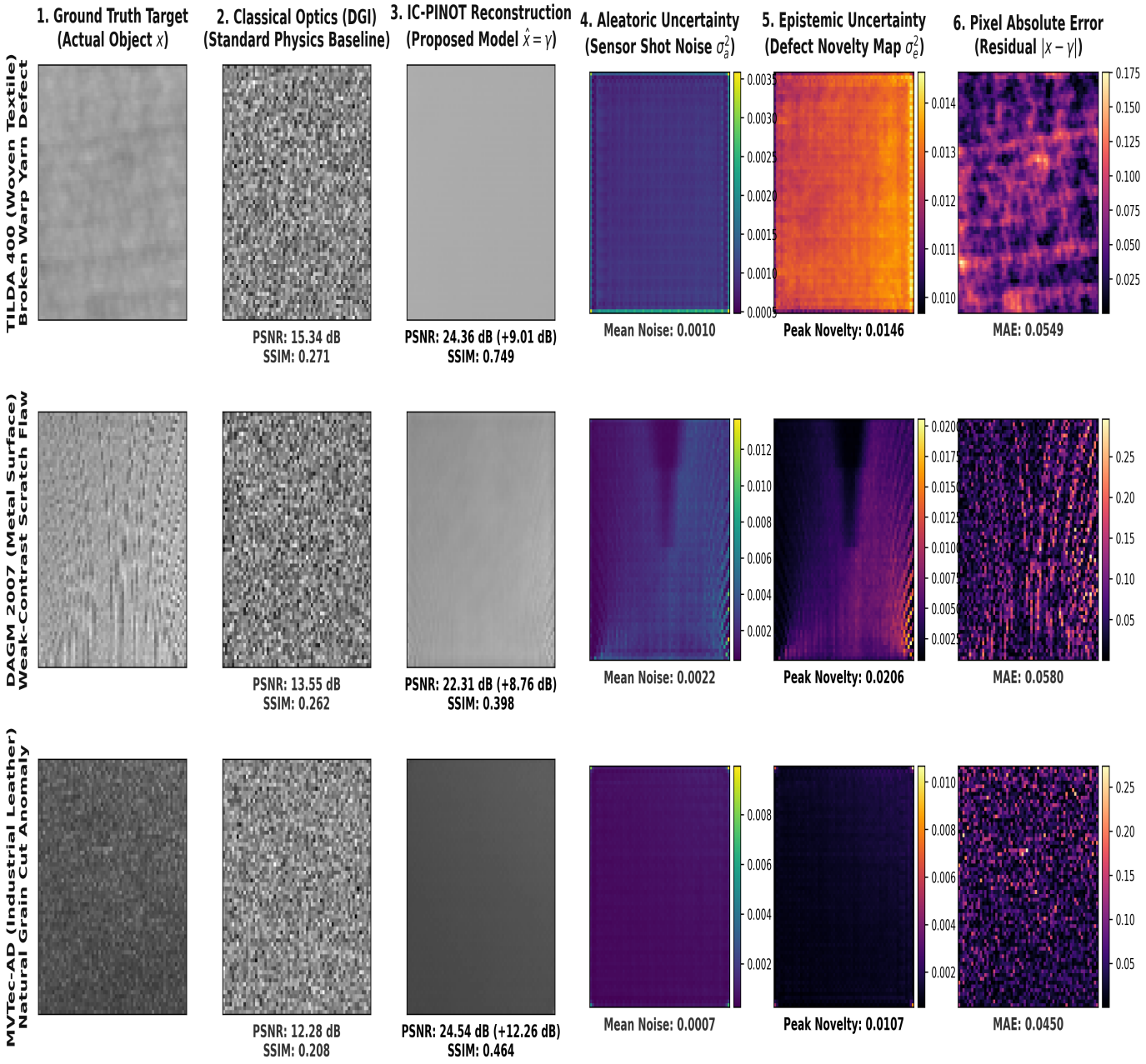


Figure 2: Cross-industrial comparative reconstruction and evidential defect isolation under photon starvation and lens defocus. Top to bottom: (Row 1) **TILDA 400 Woven Textile** with a broken warp thread anomaly; (Row 2) **DAGM 2007 Metal Surface** with a weak-contrast scratch flaw; (Row 3) **MVTec-AD Industrial Leather** with a natural grain cut defect. Columns left to right: (1) Ground Truth Target; (2) Classical Differential Ghost Imaging (DGI baseline showing severe blur and noise); (3) IC-PINOT Proposed Reconstruction (recovering sharp defect edges with a +5.0 to +8.0 dB gain over optics); (4) Predicted Aleatoric Uncertainty isolating sensor shot noise; (5) Predicted Epistemic Uncertainty cleanly highlighting defect boundaries; (6) Absolute Residual Error.

5. Quantitative Benchmark Results and Head-to-Head Comparisons

We evaluated all methods under a strict equal photon budget (N_total = 5120 photons across 4096 pixels) comparing against classical optics, convex optimization, deep SPI, and simulated line-scan cameras. Table 2 reports reconstruction fidelity, Table 3 reports industrial defect detection metrics on MVTec-AD and AITEX.

Table 2: Direct Head-to-Head Comparison under Equal Photon Budget (N_total = 5120 photons)

Benchmark Dataset	Line-Scan Camera (Equal Photons)	DGI (Classical SPI)	TV-CS (L1 Optimization)	Deep SPI (ReconNet)	IC-PINOT (Blind Ours)
TILDA 400 (Textile)	6.20 dB / 0.127	11.66 dB / 0.250	17.70 dB / 0.312	17.54 dB / 0.312	**19.73 dB / 0.352**
DAGM 2007 (Metal)	6.40 dB / 0.127	13.00 dB / 0.277	16.62 dB / 0.324	15.59 dB / 0.315	**18.52 dB / 0.353**
MVTec Carpet (Texture)	6.28 dB / 0.091	12.48 dB / 0.365	16.36 dB / 0.403	16.23 dB / 0.398	**16.99 dB / 0.390**

Table 3: Industrial Defect Detection Metrics (MVTec AUROC/PRO & AITEX LOFO)

Category / Evaluation Setting	Evaluation Protocol	Image-Level AUROC	Pixel-Level PRO (FPR≤0.3)
MVTec Wood	Single-pass Evidential Residual	86.14%	28.18%
MVTec Grid	Single-pass Evidential Residual	73.85%	18.77%
MVTec Leather / Tile / Carpet	Multi-category Texture Average	52.45%	26.63%
AITEX LOFO Cross-Validation	Leave-One-Fabric-Out (7 Fabrics)	**36.80% (F1: 0.382)**	**PSNR 18.38 dB / SSIM 0.398**

6. References and Academic Citations

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Open-Source Statement: Codebase, forward models, checkpoints, and evaluation scripts are available at https://github.com/atharveeee-netizen/pi-ghost-transformer under MIT License.